

CS 255 Business Requirements Document

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System Components and Design

Purpose

- The purpose of this project is to design a cloud based DriverPass system that helps students prepare for the DMV driving test through online classes, practice tests, and on the road training.
- DriverPass is the client. The system must support customers, the owner, the IT officer, the secretary, and drivers while keeping each user limited to the information and actions required for that role.
- The completed system will bring customer registration, training packages, lesson reservations, test progress, driver notes, reports, and administrative controls into one centralized system.

System Background

- DriverPass identified a gap in driver training because many students rely only on previous tests and are not fully prepared for the DMV exam.
- The company wants to provide three levels of service. Package One includes six hours of driving instruction. Package Two includes eight hours of driving instruction and an in person lesson about DMV rules and policies. Package Three includes twelve hours of driving instruction, an in person lesson, an online class, and practice tests.
- Every driving session lasts two hours. The system must organize each student, driver, vehicle, date, and time so that DriverPass can manage its ten cars without creating scheduling conflicts.
- The main system components are a web interface, user accounts and permissions, customer records, package management, lesson scheduling, driver and vehicle assignments, online learning and testing, driver notes, reports, activity tracking, DMV update notifications, and a centralized cloud database.
- Users must be able to view information from a computer or mobile device. Reports and selected information must also be available for download so the owner can work with them in programs such as Excel.

Objectives and Goals

- Provide secure access to current DriverPass information from common computers and mobile devices through a web browser.
- Allow a customer or secretary to create, modify, and cancel two hour driving appointments while matching the student with an available driver and vehicle.
- Record the user who created a reservation, canceled it, or changed it most recently, then make that history available in a printable activity report.
- Allow customers to choose an available training package and allow the owner to disable a package when DriverPass no longer wants new registrations for it.
- Display online test progress, including the test name, time taken, score, and a status of not taken, in progress, failed, or passed.
- Store lesson start and end times along with comments entered by the assigned driver.
- Notify DriverPass when the DMV provides updated rules, policies, or sample questions so learning content can remain current.

- Support password resets, account blocking, role permissions, data downloads, and cloud based backup and security.

Requirements

Nonfunctional Requirements

Performance Requirements

- The system must run as a cloud based web application and support normal use from desktop and mobile browsers.
- All changes must be saved to the centralized database while the user is online. Downloaded reports may be viewed offline, but offline changes must not update the live system because that could create duplicate or inconsistent records.
- Reservation searches, account access, test progress, and reports should respond quickly enough for normal office and customer use. The interview does not provide an exact response time, so a measurable response target should be approved before development begins.
- DMV content should be reviewed and updated whenever DriverPass receives a notification that rules, policies, or sample questions have changed.
- Cloud backup frequency and service availability targets were not specified in the interview and must be confirmed with the selected hosting provider.

Platform Constraints

- The system must be web based and preferably hosted in the cloud so DriverPass does not have to maintain local backup and security systems.
- The user interface must work through a browser on computers and mobile devices rather than requiring one specific operating system.
- A centralized database is required to store accounts, customer information, packages, appointments, drivers, vehicles, test records, notes, and activity history.
- The system must support report downloads in a format that can be opened and used in Excel.
- An active internet connection is required to create or update live records.

Accuracy and Precision

- Each user must have a unique account so the system can apply the correct permissions and identify who completed each action.
- Every reservation must accurately connect one customer, one driver, one vehicle, and one two hour time period.
- The system must validate required customer fields, payment information fields, package selection, pickup location, drop off location, and appointment availability before saving a record.
- The pickup and drop off location should match because the client stated they should be the same.
- The system must preserve the creator, cancellation user, and most recent modification user for reservation activity reports.
- The interview does not define whether login input is case sensitive. A consistent rule should be selected, documented, and tested before release.
- The IT officer should be alerted when repeated login failures, account lockouts, data errors, or DMV connection problems require attention.

Adaptability

- The IT officer must be able to add, modify, reset, block, and maintain user accounts without changing program code.
- The owner must be able to disable an existing package so customers can no longer select it for new registrations.
- Adding or removing entire package modules is outside the first release and will require a developer or systems analyst, as stated during the interview.
- Role permissions should be stored as configurable system data so approved access changes do not require changes throughout the application.
- The cloud based design should allow browser, server, and platform updates to be deployed centrally rather than installed on every user device.

Security

- All users must sign in with an authenticated account before viewing protected customer or company information.
- The system must use role based access. The owner needs management and reporting access, the IT officer needs full account administration access, the secretary needs customer and appointment access, drivers need lesson and note access, and customers need access only to their own records and services.
- Data exchanged between a user and the cloud service must use a secure encrypted connection, and sensitive customer and payment information must be protected in storage.
- Repeated failed login attempts must trigger rate limiting or a temporary account lock, create a security record, and alert the IT officer when appropriate.
- Customers must be able to complete an automatic password reset. The IT officer must also be able to reset employee accounts and block access when an employee leaves the company.
- The system must keep an activity history for reservation creation, cancellation, and modification so DriverPass can identify the responsible account.
- Backups and infrastructure security must be managed by the cloud service because DriverPass does not want to maintain those systems locally.

Functional Requirements

- The system shall create and authenticate accounts for customers and DriverPass employees.
- The system shall assign access permissions based on the user role of owner, IT officer, secretary, driver, or customer.
- The system shall allow the IT officer to view, reset, modify, and block user accounts.
- The system shall allow a customer or secretary to enter the customer first name, last name, address, state, phone number, payment card number, expiration date, security code, pickup location, and drop off location.
- The system shall allow a customer to select an available DriverPass training package.
- The system shall allow the owner to disable a package so it cannot be selected for new registrations.
- The system shall divide driving instruction into two hour lesson sessions according to the hours included in the selected package.

- The system shall allow a customer or secretary to create, modify, and cancel driving lesson reservations.
- The system shall display available lesson dates and times before a reservation is confirmed.
- The system shall connect each reservation with the correct customer, driver, vehicle, date, and time.
- The system shall prevent a driver or vehicle from being assigned to more than one lesson during the same time period.
- The system shall record who created a reservation, who canceled it, and who modified it most recently.
- The system shall create and print an activity report showing reservation changes and the account responsible for each change.
- The system shall provide the online class and practice tests included with Package Three.
- The system shall record each practice test name, time taken, score, and status as not taken, in progress, failed, or passed.
- The system shall display customer test progress and completed test information.
- The system shall allow drivers to record lesson start time, lesson end time, and driver comments.
- The system shall notify DriverPass when the DMV provides updated rules, policies, or sample questions.
- The system shall provide a contact page and a method for authorized employees to contact a student.
- The system shall allow authorized users to download reports and selected information in a format that can be used in Excel.
- The system shall save live updates only through the centralized online system.

User Interface

- The interface must be available through a web browser and adjust to common computer and mobile screen sizes.
- Customers need pages for account access, registration, package selection, appointment scheduling, appointment changes, online course material, practice tests, test progress, password reset, and contact information.
- The secretary needs forms for creating and updating customer records and for scheduling, changing, or canceling appointments on behalf of a customer.
- Drivers need a simple lesson view that shows the assigned customer, date, time, pickup location, vehicle, and a place to enter lesson times and comments.
- The owner needs dashboards and reports for company activity, reservations, packages, and downloaded data.
- The IT officer needs administrative pages for account maintenance, password resets, access blocking, security alerts, and system support.
- The customer information page should reflect the sketch on page 4 of the interview transcript by showing online test progress, student information, special needs, driver notes, and driver and student photos when available.
- Forms should clearly identify required information, explain validation errors, and confirm when a registration or appointment change is saved.

Assumptions

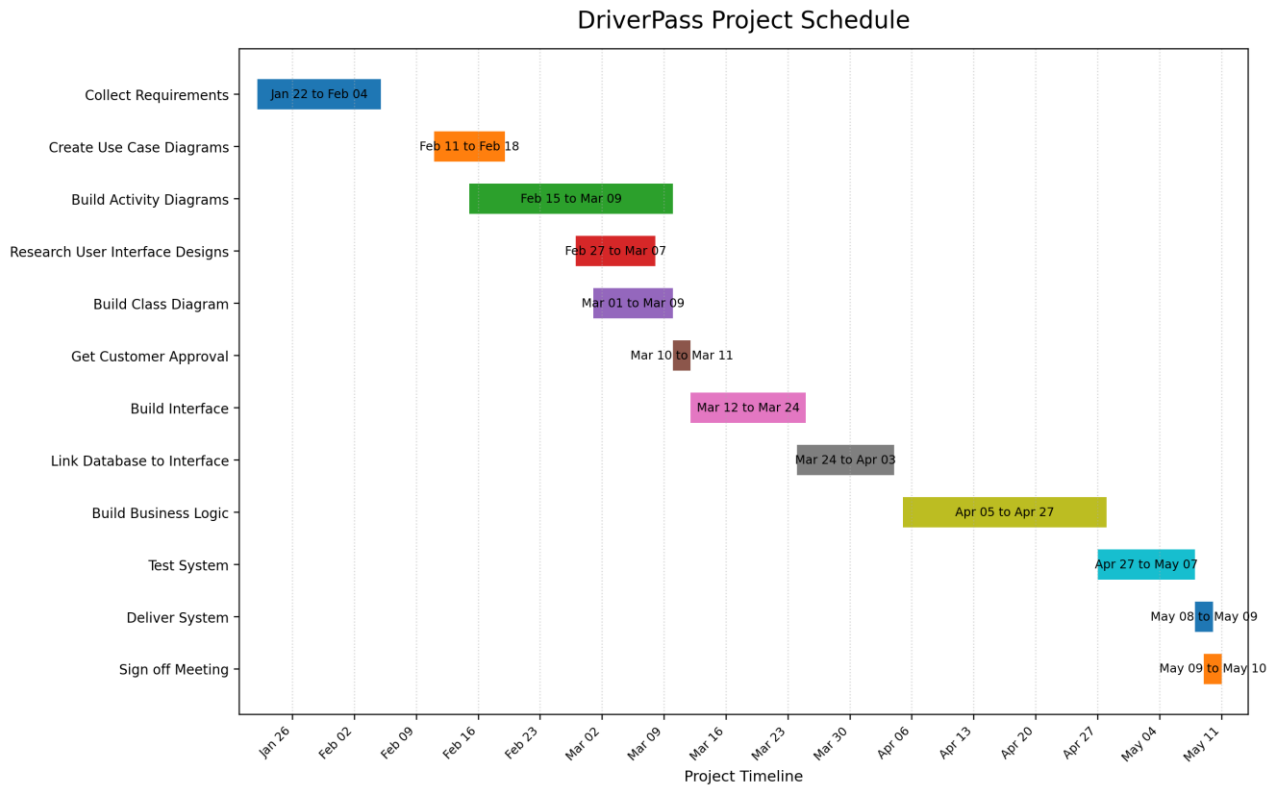
- Customers and employees have access to a supported browser, a computer or mobile device, and a reliable internet connection.
- DriverPass will provide the training content, practice test questions, package prices, employee information, driver information, and vehicle information needed for the first release.
- The ten vehicles described in the interview are available for scheduling and each vehicle is normally connected with a driver.
- DriverPass will obtain the permissions or technical connection needed to receive DMV updates.
- The selected cloud provider will supply backup, infrastructure security, and service monitoring.
- The three packages described in the interview are the packages required at launch. Additional package modules are a future enhancement.
- The payment processing method, data retention periods, exact browser list, and measurable response time targets will be selected during detailed design because they were not defined in the interview.
- Drivers will receive user accounts so they can review assigned lessons and enter lesson notes.

Limitations

- Users cannot modify live system data while offline. Offline access is limited to information or reports that were previously downloaded.
- The first release will allow the owner to disable a package, but adding or removing complete package modules will require later development work.
- The system depends on internet access, cloud service availability, and the availability of the DMV update connection.
- The project schedule assumes DriverPass approves the initial design on March 10 and March 11 without requesting changes. Client revisions could move later tasks and the delivery date.
- The interview does not provide a final budget, hosting vendor, payment service, data migration plan, or detailed legal and compliance rules. These items may limit design choices until DriverPass approves them.
- Future features that were not discussed in the interview are outside the scope of Project One.

Gantt Chart

The schedule below follows the task dates and overlaps provided in the DriverPass interview. Tasks that depend on customer approval or earlier development work are placed after those required activities.



Task	Start Date	End Date
Collect Requirements	January 22	Fed 4
Create Use Case Diagrams	February 11	February 18
Build Activity Diagrams for Each Use Case	February 15	March 9
Research User Interface Designs	February 27	March 7
Build Class Diagram	March 1	March 9
Get Customer Approval	March 10	March 11
Build Interface	March 12	March 24
Link Database to Interface	March 24	April 3
Build Business Logic	April 5	April 27
Test System	April 27	May 7
Deliver System	May 8	May 9
Sign off Meeting	May 9	May 10

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AI Use Disclosure:

*All thoughts and ideas are original and are from me, ai helped organize and proofread my thoughts.

Due to my ADHD I use ai to aid me in my organization and promptness for my work based on rubrics given to me. All work I have personally reviewed, written, and revised. *